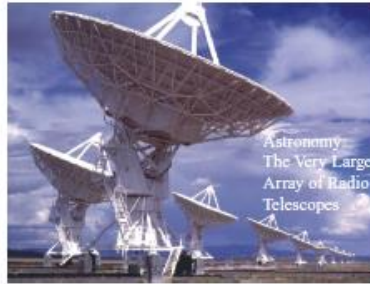
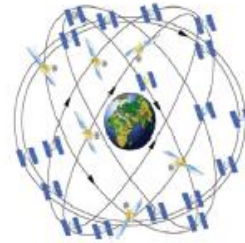


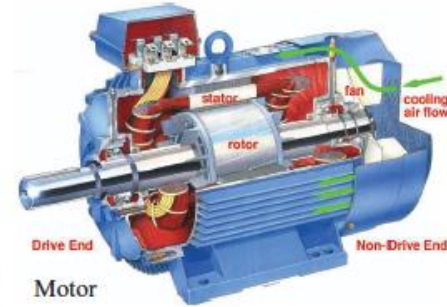
Examples of EM Applications



Astronomy
The Very Large
Array of Radio
Telescopes

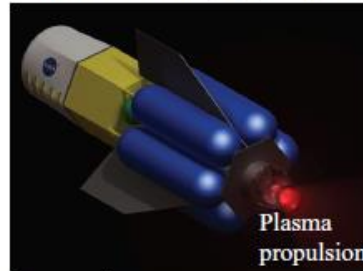


Global Positioning System (GPS)

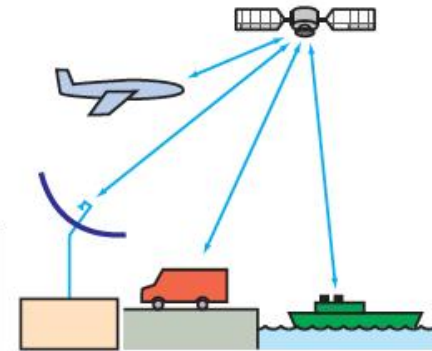


Motor

LCD
Screen



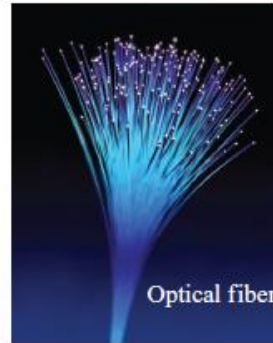
Plasma
propulsion



Telecommunication



Radar

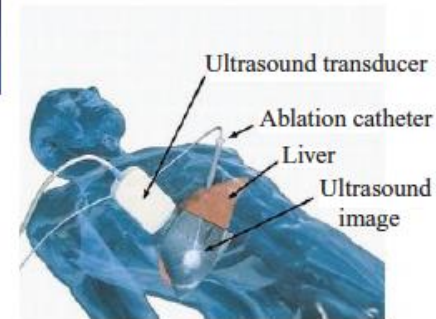


Optical fiber



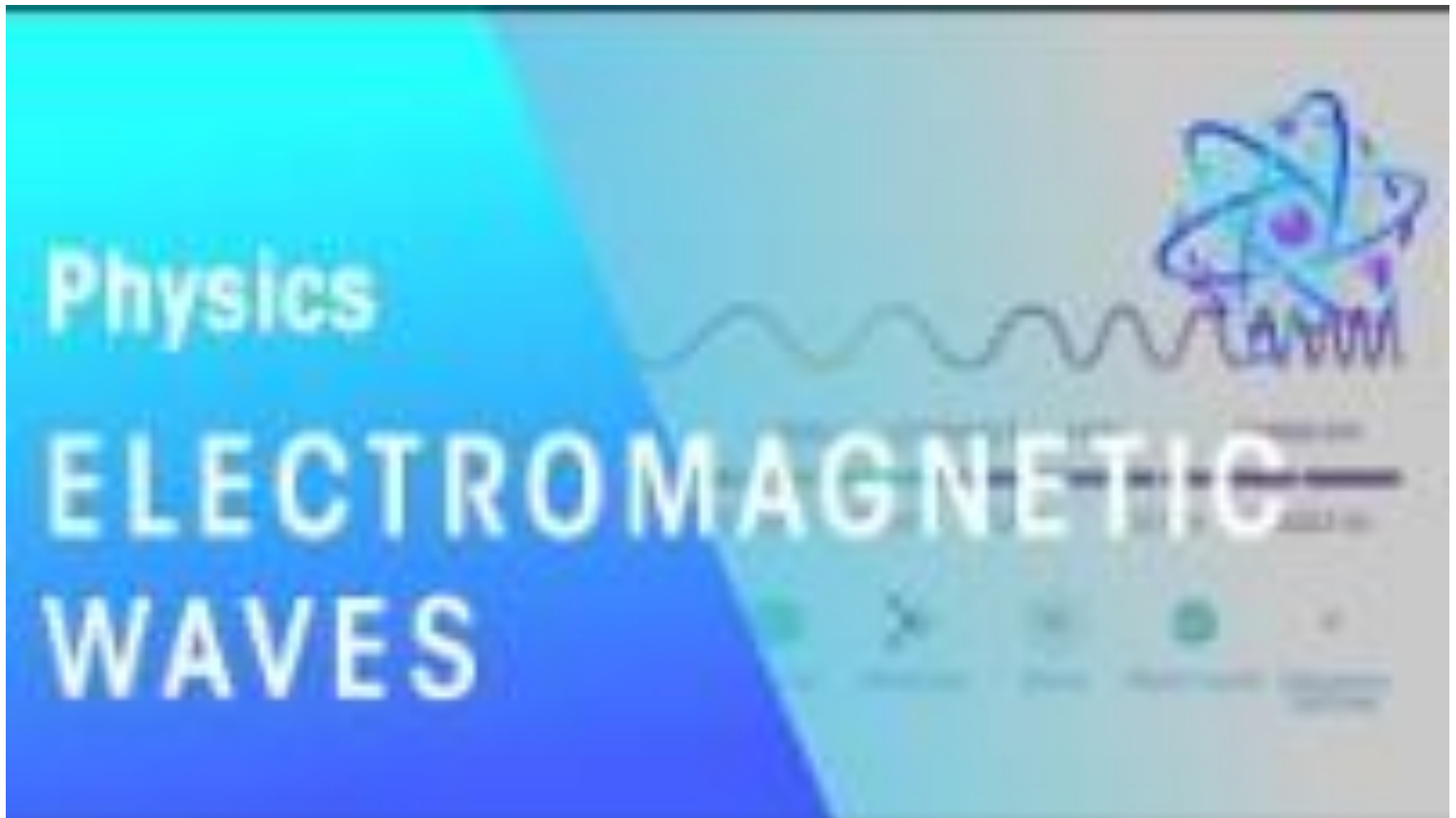
Electromagnetic sensors

Cell
phone



Microwave ablation for
liver cancer treatment

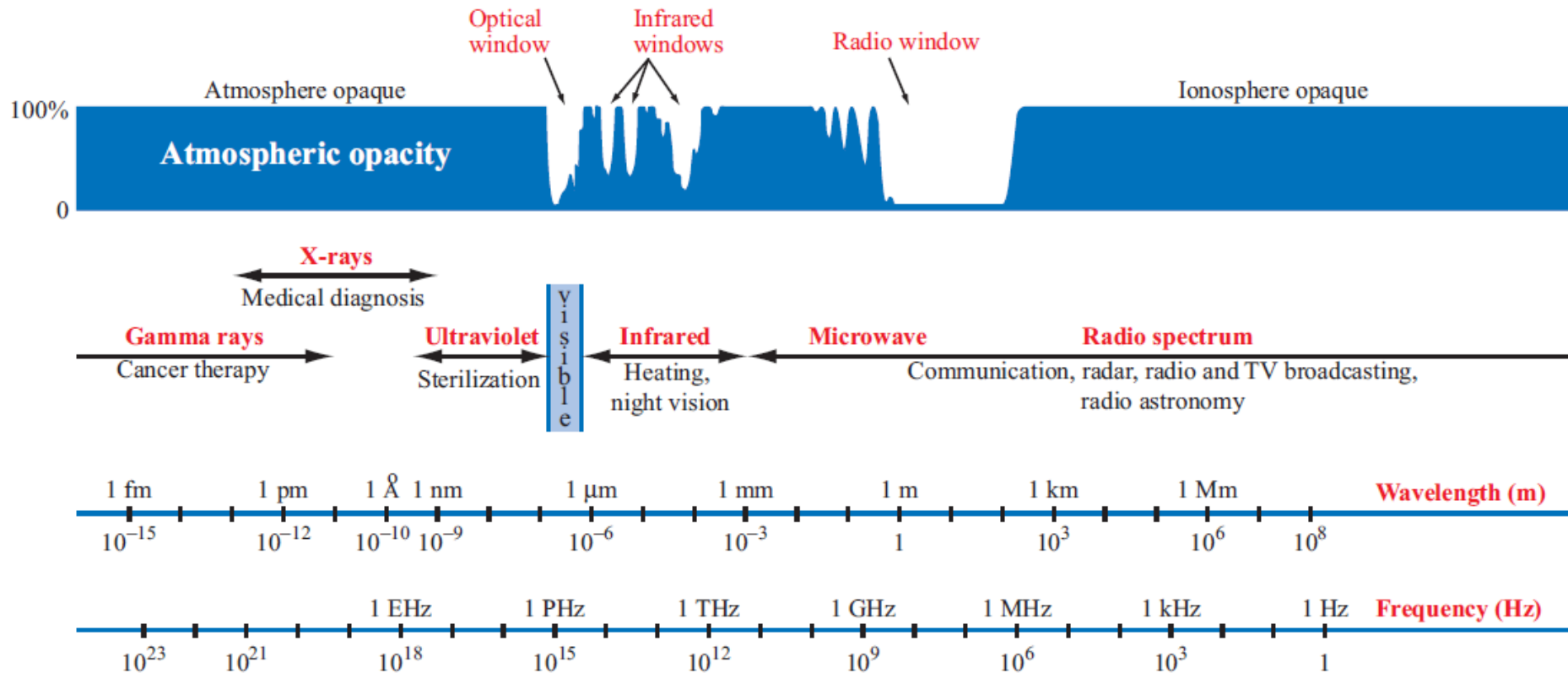
Electromagnetic Waves Applications



<https://www.youtube.com/watch?v=508ZsmsIlno>

The EM Spectrum

$$\lambda = \frac{c}{f}$$



UNITED STATES

FREQUENCY ALLOCATIONS THE RADIO SPECTRUM

RADIO SERVICES COLOR LEGEND

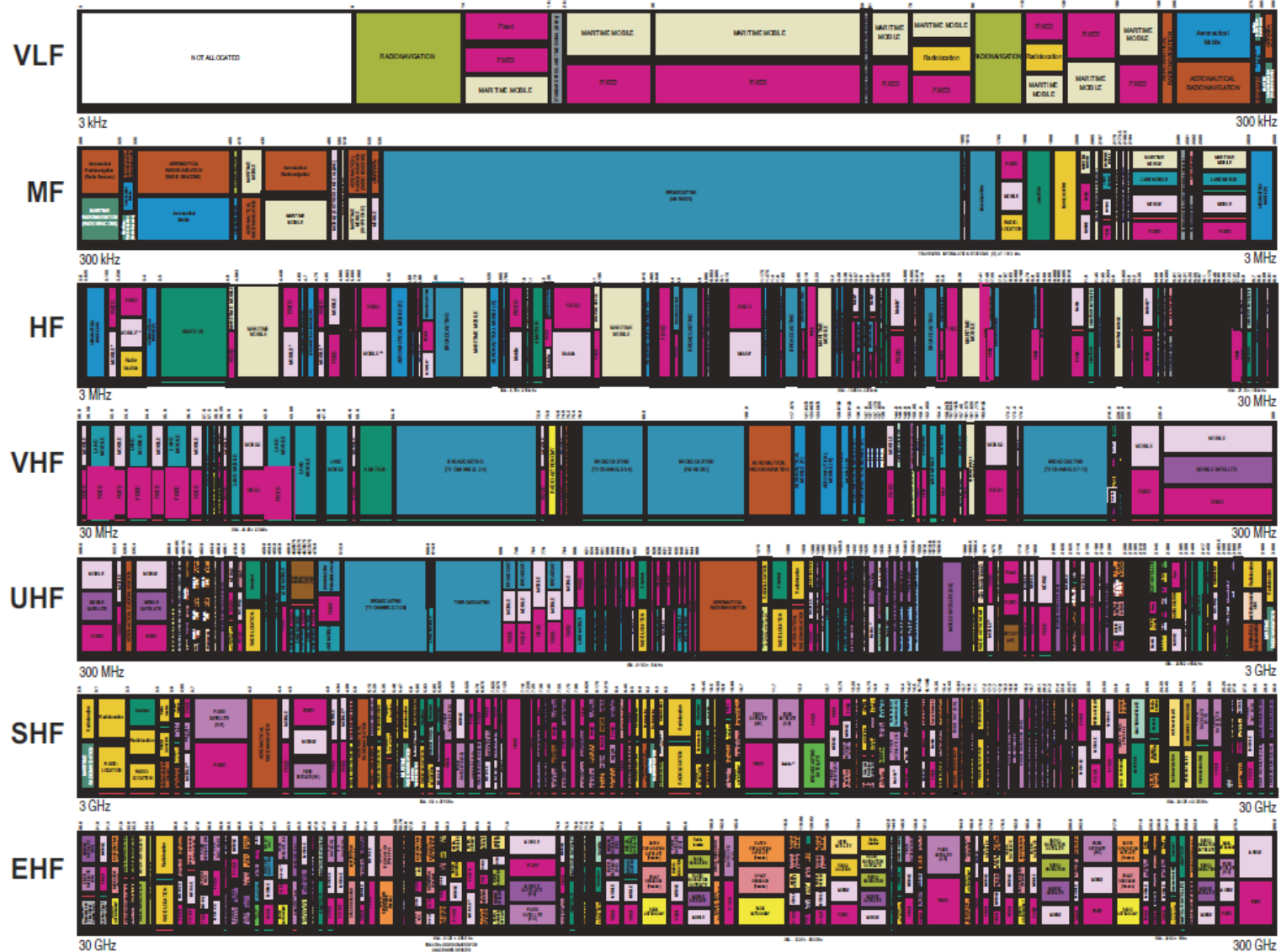
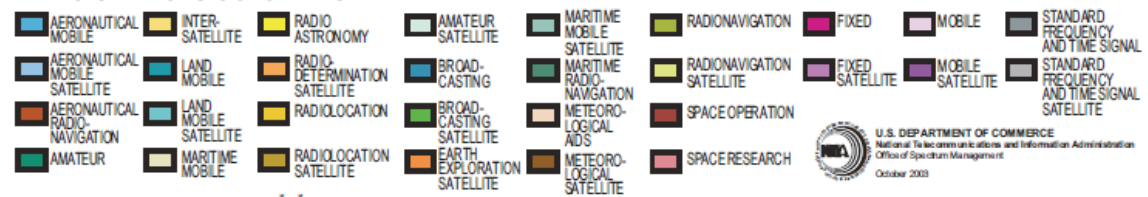


Figure 1-17: Individual bands of the radio spectrum and their primary allocations in the US. [See expandable version on CD.]



Use ChatGPT to explore the following



What is a wave?



What is special about electromagnetic waves?

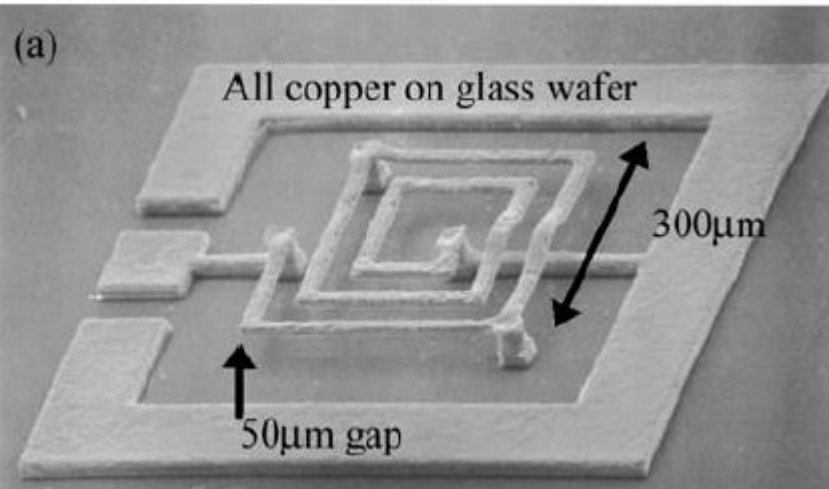


What is a sinusoidal solution of wave equation?

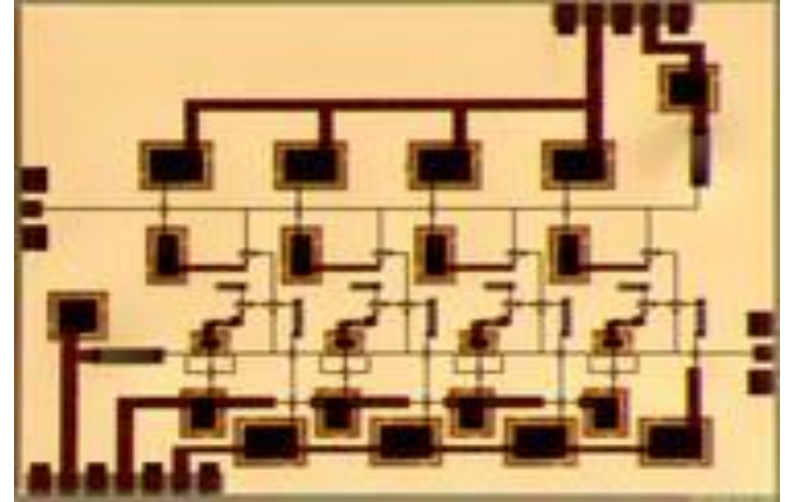


What does an RF Engineer (mm-Wave Engineer) do?

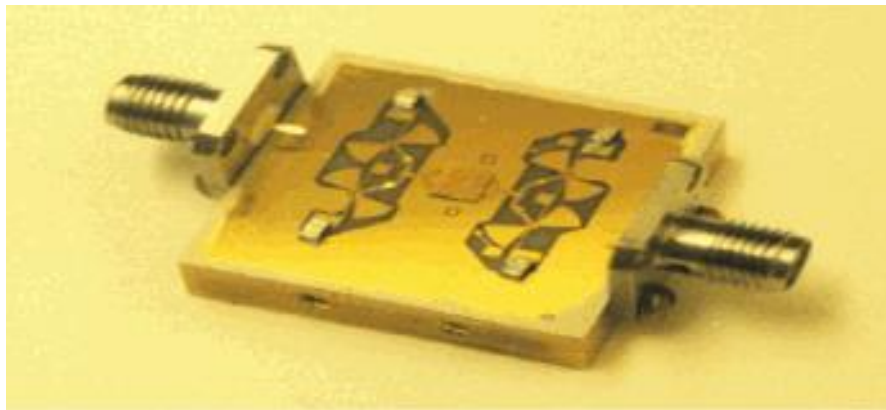
Example of Microwave Circuits



A Planar Chip Inductor



Traveling-wave amplifier



GaN Amplifier for 10 GHz



Microwave Mixer

Example of Microwave Antennas

1 m Dipole Antenna TV



2 m Satellite Receiver



High Gain
Dish Antenna



Antenna For
Mobile Systems



Dimensions and Units

Table 1-1: Fundamental SI units.

Dimension	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol

Table 1-2: Multiple and submultiple prefixes.

Prefix	Symbol	Magnitude
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}

Electromagnetics in the Classical Era

1827 **Georg Simon Ohm** (German) formulates Ohm's law relating electric potential to current and resistance.

1827 **Joseph Henry** (American) introduces the concept of **inductance**, and builds one of the earliest electric motors. He also assisted Samuel Morse in the development of the **telegraph**.

1831 **Michael Faraday** (English) discovers that a changing magnetic flux can induce an **electromotive force**.



1835 **Carl Friedrich Gauss** (German) formulates **Gauss's law** relating the electric flux flowing through an enclosed surface to the enclosed electric charge.

Gauss's Law for Electricity

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{inside}}}{\epsilon_0}$$

1873 **James Clerk Maxwell** (Scottish) publishes his Treatise on Electricity and Magnetism in which he unites the discoveries of Coulomb, Oersted, Ampère, Faraday, and others into four elegantly constructed mathematical equations, now known as **Maxwell's Equations**.



1887 **Heinrich Hertz** (German) builds a system that can generate **electromagnetic waves** (at radio frequencies) and detect them.



1888 **Nikola Tesla** (Croatian-American) invents the **ac** (alternating current) electric motor.



1895 **Wilhelm Röntgen** (German) discovers **X-rays**. One of his first X-ray images was of the bones in his wife's hands. [1901 Nobel prize in physics.]



1897 **Joseph John Thomson** (English) discovers the **electron** and measures its charge-to-mass ratio. [1906 Nobel prize in physics.]

1905 **Albert Einstein** (German-American) explains the **photoelectric effect** discovered earlier by Hertz in 1887. [1921 Nobel prize in physics.]

Telecommunications

1923



Vladimir Zworykin (Russian-American) invents **television**. In 1926, John Baird (Scottish) transmits TV images over telephone wires from London to Glasgow. Regular **TV broadcasting** began in Germany (1935), England (1936), and the United States (1939).

1958

Jack Kilby (American) builds first **integrated circuit (IC)** on germanium and, independently, **Robert Noyce** (American) builds first IC on silicon.



1926

Transatlantic telephone service between London and New York.

1932

First **microwave telephone link**, installed (by Marconi) between Vatican City and the Pope's summer residence.

1933

Edwin Armstrong (American) invents **frequency modulation (FM)** for radio transmission.

1935

Robert Watson-Watt (Scottish) invents **radar**.



1938

H. A. Reeves (American) invents **pulse code modulation (PCM)**.

1947

William Shockley, Walter Brattain, and John Bardeen (all Americans) invent the junction **transistor** at Bell Labs. [1956 Nobel prize in physics.]



1960



Echo, the first passive communication satellite is launched, and successfully reflects radio signals back to Earth. In 1963, the first **communication satellite** is placed in geosynchronous orbit.

1969

ARPANET is established by the U.S. Department of Defense, to evolve later into the Internet.

1979

Japan builds the first cellular telephone network:

- 1983 cellular phone networks start in the United States.
- 1990 **electronic beepers** become common.
- 1995 **cell phones** become widely available.
- 2002 cell phone supports **video and Internet**.

1984

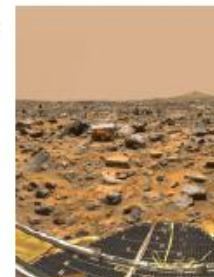
Worldwide **Internet** becomes operational.

1988

First **transatlantic optical fiber cable** between the U.S. and Europe.

1997

Mars Pathfinder sends images to Earth.



1955

Pager is introduced as a radio communication product in hospitals and factories.

1955

Narinder Kapany (Indian-American) demonstrates the **optical fiber** as a low-loss, light-transmission medium.

2004

Wireless communication supported by many airports, university campuses, and other facilities.

2012

Smartphones worldwide exceed 1 billion.

Computer Technology

1968 **Douglas Engelbart** (American) demonstrates a **word-processor** system, the mouse pointing device and the use of "**windows**."

1971 **Texas Instruments** introduces the **pocket calculator**.



1971 **Ted Hoff** (American) invents the Intel 4004, the first computer **microprocessor**.

1976 **IBM** introduces the **laser printer**.

1976 **Apple Computer** sells **Apple I** in kit form, followed by the fully assembled **Apple II** in 1977 and the **Macintosh** in 1984.



1980 **Microsoft** introduces the **MS-DOS** computer disk operating system. Microsoft Windows is marketed in 1985.



1981 **IBM** introduces the **PC**.

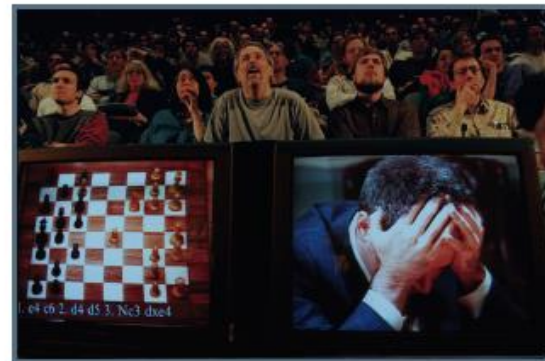
1989 **Tim Berners-Lee** (British) invents the **World Wide Web** by introducing a networked hypertext system.

1991 **Internet** connects to 600,000 hosts in more than 100 countries.

1995 **Sun Microsystems** introduces the **Java** programming language.

1996 **Sabeer Bhatia** (Indian-American) and **Jack Smith** (American) launch **Hotmail**, the first webmail service.

1997 **IBM's Deep Blue** computer defeats World Chess Champion Garry Kasparov.



2002 The billionth **personal computer** was sold, second billion reached in 2007.

2010 **iPad** introduced in 2010.



Material Properties

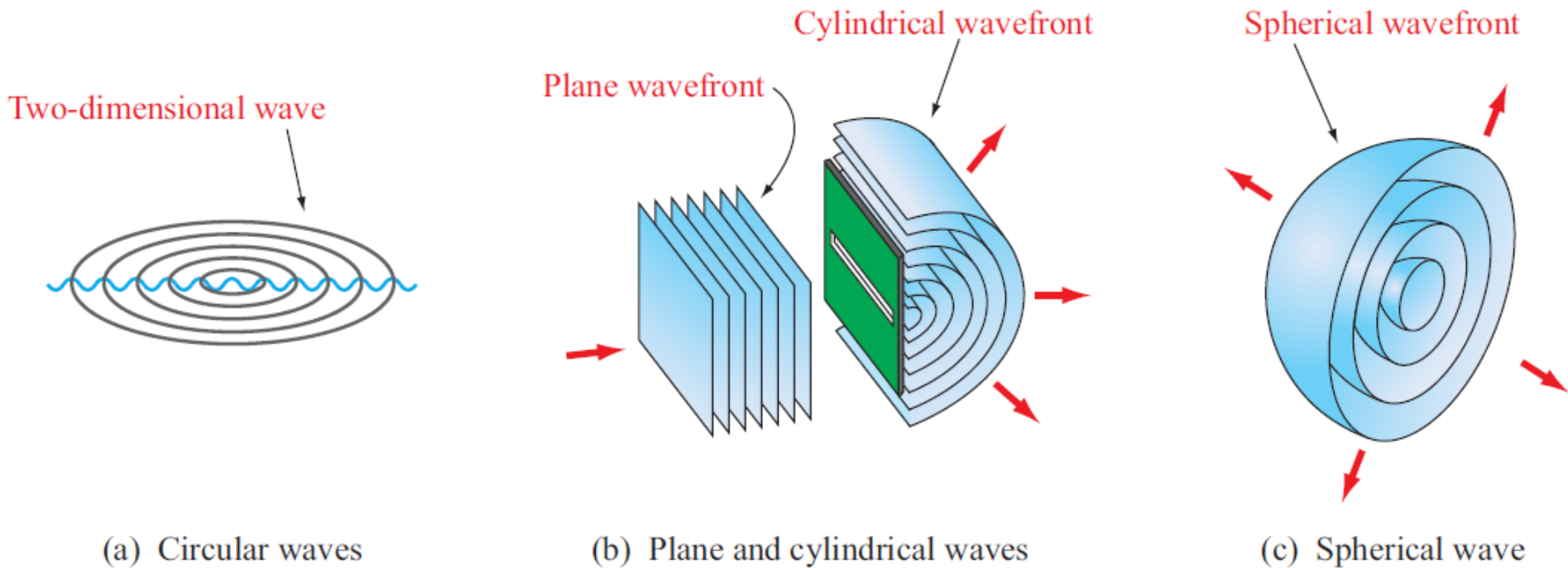
Table 1-4: Constitutive parameters of materials.

Parameter	Units	Free-space Value
Electrical permittivity ε	F/m	$\varepsilon_0 = 8.854 \times 10^{-12} \text{ (F/m)}$ $\simeq \frac{1}{36\pi} \times 10^{-9} \text{ (F/m)}$
Magnetic permeability μ	H/m	$\mu_0 = 4\pi \times 10^{-7} \text{ (H/m)}$
Conductivity σ	S/m	0

Traveling Waves

- Waves carry energy
- Waves have velocity
- Many waves are linear: they do not affect the passage of other waves; they can pass right through them
- **Transient waves:** caused by sudden disturbance
- **Continuous periodic waves:** repetitive source

Types of Waves



Sinusoidal Waves in Lossless Media

A medium is said to be **lossless** if it does not attenuate the amplitude of the wave traveling within it or on its surface.

y = height of water surface

x = distance

$$y(x, t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \text{ (m)}, \quad (1.17)$$

where A is the **amplitude** of the wave, T is its **time period**, λ is its **spatial wavelength**, and ϕ_0 is a **reference phase**.

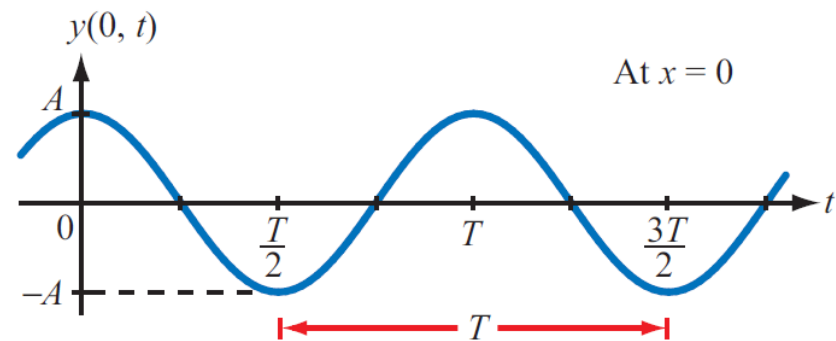
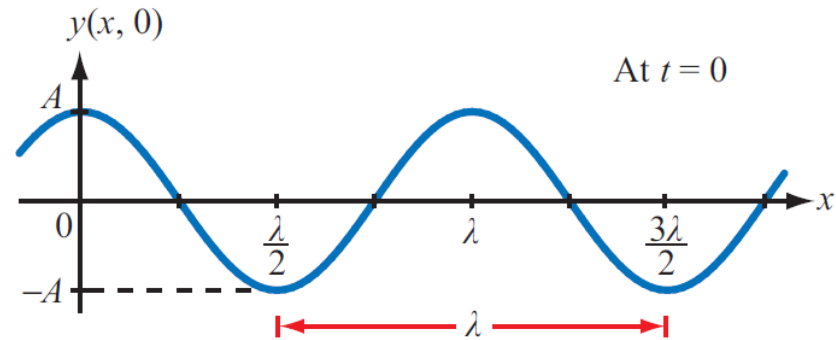


Figure 1-12: Plots of $y(x, t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right)$ as a function of (a) x at $t = 0$ and (b) t at $x = 0$.

Phase velocity

$$y(x, t) = A \cos \phi(x, t) \quad (\text{m}),$$

where

$$\phi(x, t) = \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \quad (\text{rad}).$$

If we select a fixed height y_0 and follow its progress, then

$$y_0 = y(x, t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right)$$

$$\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = \cos^{-1} \left(\frac{y_0}{A} \right) = \text{constant}$$

$$\frac{2\pi}{T} - \frac{2\pi}{\lambda} \frac{dx}{dt} = 0$$

$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad (\text{m/s})$$

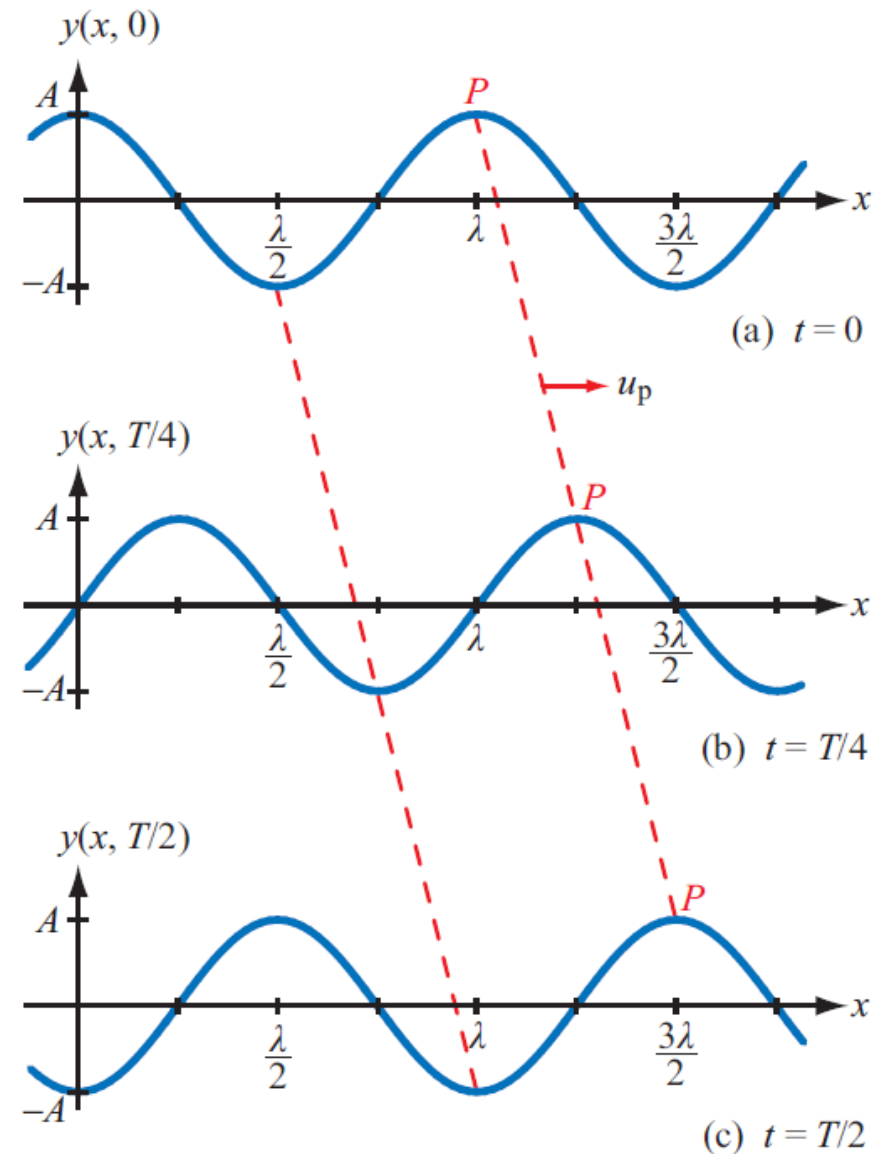


Figure 1-13: Plots of $y(x, t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right)$ as a function of x at (a) $t = 0$, (b) $t = T/4$, and (c) $t = T/2$. Note that the wave moves in the $+x$ -direction with a velocity $u_p = \lambda/T$.

Wave Frequency and Period

The *frequency* of a sinusoidal wave, f , is the reciprocal of its time period T :

$$f = \frac{1}{T} \quad (\text{Hz}). \quad (1.26)$$

Combining the preceding two equations yields

$$u_p = f\lambda \quad (\text{m/s}). \quad (1.27)$$

Using Eq. (1.26), Eq. (1.20) can be rewritten in a more compact form as

$$y(x, t) = A \cos \left(2\pi f t - \frac{2\pi}{\lambda} x \right) = A \cos(\omega t - \beta x), \quad (1.28)$$

where ω is the *angular velocity* of the wave and β is its *phase constant* (or *wavenumber*), defined as

$$\omega = 2\pi f \quad (\text{rad/s}), \quad (1.29a)$$

$$\beta = \frac{2\pi}{\lambda} \quad (\text{rad/m}). \quad (1.29b)$$

Direction of Wave Travel

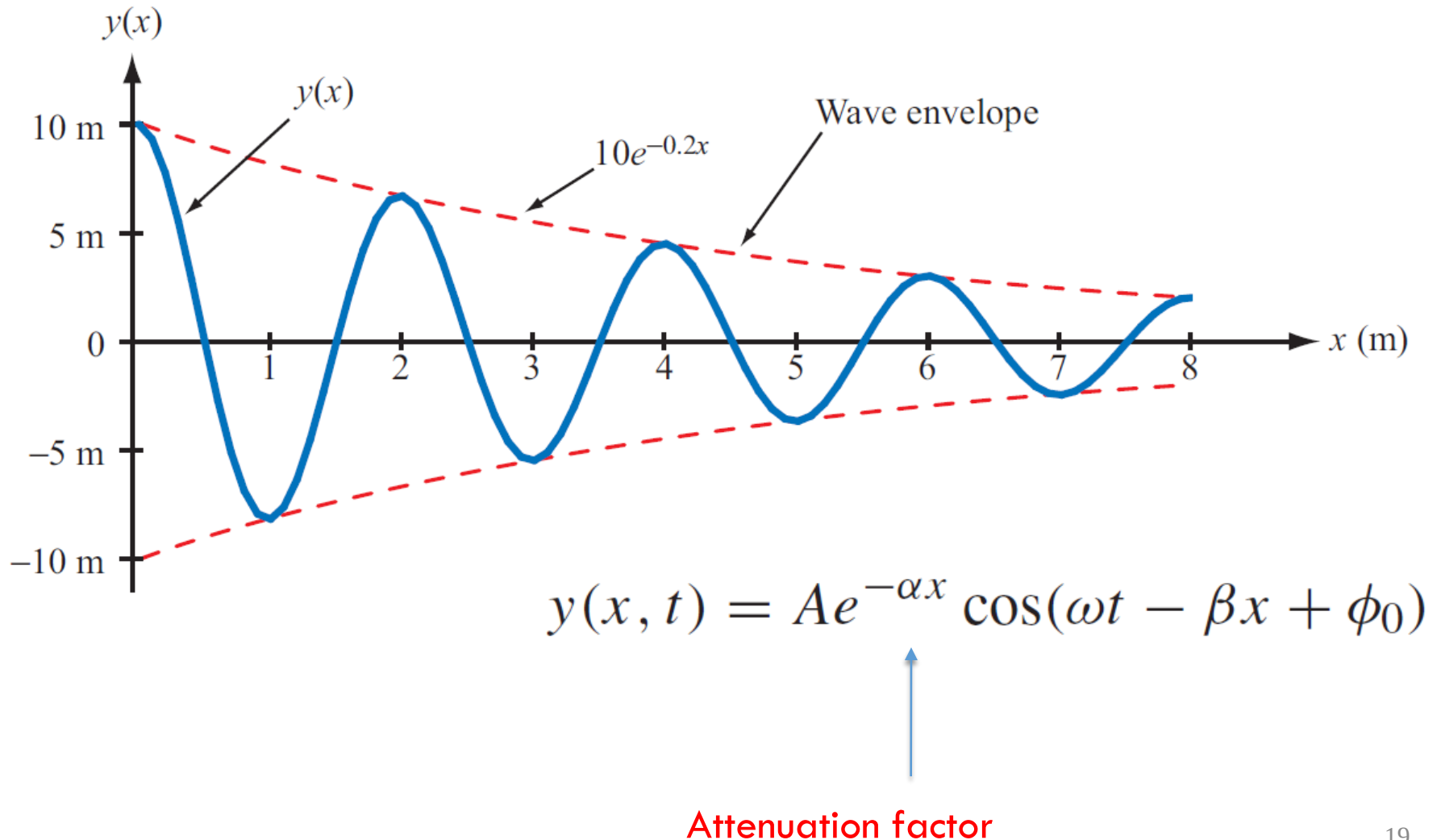
$y(x, t) = A \cos(\omega t - \beta x)$ ← Wave travelling in **+x** direction

$y(x, t) = A \cos(\omega t + \beta x)$ ← Wave travelling in **-x** direction

+x direction: if coefficients of t and x have opposite signs

-x direction: if coefficients of t and x have same sign (both positive or both negative)

Wave Travel in Lossy Media



Example 1-1: Sound Wave in Water

Given: sinusoidal sound wave traveling in the positive x -direction in water

Wave amplitude is 10 N/m^2 , and $p(x, t)$ was observed to be at its maximum value at $t = 0$ and $x = 0.25 \text{ m}$. Also $f = 1 \text{ kHz}$, $u_p = 1.5 \text{ km/s}$.

Determine: $p(x, t)$

Solution:

$$p(x, t) = A \cos\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x + \phi_0\right) \quad (\text{N/m}^2).$$

The amplitude $A = 10 \text{ N/m}^2$, $T = 1/f = 10^{-3} \text{ s}$, and from $u_p = f\lambda$,

$$\begin{aligned}\lambda &= \frac{u_p}{f} \\ &= \frac{1.5 \times 10^3}{10^3} \\ &= 1.5 \text{ m}.\end{aligned}$$

Hence,

$$p(x, t) = 10 \cos\left(2\pi \times 10^3 t - \frac{4\pi}{3}x + \phi_0\right) \quad (\text{N/m}^2).$$

Since at $t = 0$ and $x = 0.25 \text{ m}$, $p(0.25, 0) = 10 \text{ N/m}^2$, we have

$$10 = 10 \cos\left(\frac{-4\pi}{3} \cdot 0.25 + \phi_0\right)$$

$$= 10 \cos\left(\frac{-\pi}{3} + \phi_0\right),$$

which yields the result $(\phi_0 - \pi/3) = \cos^{-1}(1)$, or $\phi_0 = \pi/3$. Hence,

$$p(x, t) = 10 \cos\left(2\pi \times 10^3 t - \frac{4\pi}{3}x + \frac{\pi}{3}\right) \quad (\text{N/m}^2).$$

Example 1-2: Power Loss

A laser beam of light propagating through the atmosphere is characterized by an electric field given by

$$E(x, t) = 150e^{-0.03x} \cos(3 \times 10^{15}t - 10^7x) \quad (\text{V/m}),$$

where x is the distance from the source in meters. The attenuation is due to absorption by atmospheric gases. Determine:

- (a) the direction of wave travel,
- (b) the wave velocity, and
- (c) the wave amplitude at a distance of 200 m.

Solution: (a) Since the coefficients of t and x in the argument of the cosine function have opposite signs, the wave must be traveling in the $+x$ -direction.

(b)

$$\begin{aligned} u_p &= \frac{\omega}{\beta} \\ &= \frac{3 \times 10^{15}}{10^7} \\ &= 3 \times 10^8 \text{ m/s}, \end{aligned}$$

which is equal to c , the velocity of light in free space.

(c) At $x = 200$ m, the amplitude of $E(x, t)$ is

$$150e^{-0.03 \times 200} = 0.37 \quad (\text{V/m}).$$

Complex Numbers

We will find it is useful to represent sinusoids as complex numbers

$$j = \sqrt{-1}$$

$$z = x + jy$$

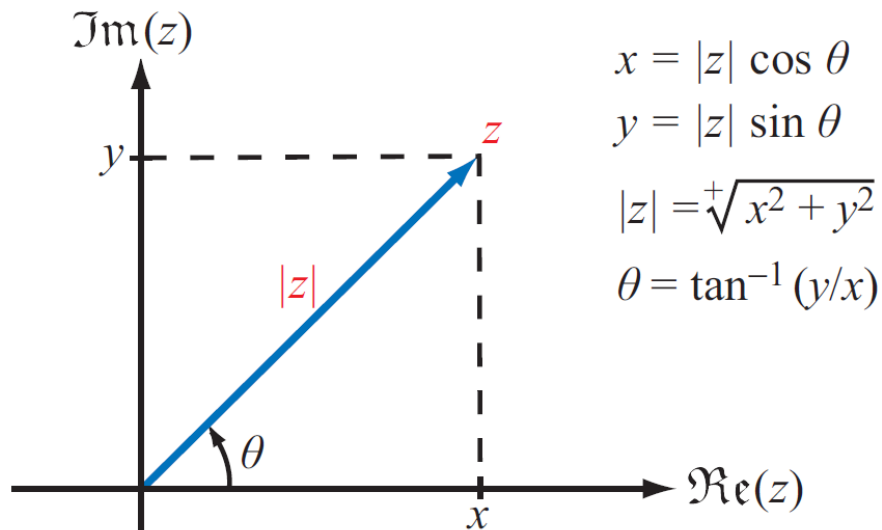
Rectangular coordinates

$$\operatorname{Re}(z) = x$$

$$z = |z| \angle \theta = |z| e^{j\theta}$$

Polar coordinates

$$\operatorname{Im}(z) = y$$



**Relations based
on Euler's Identity**

$$e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

Relations for Complex Numbers

Euler's Identity: $e^{j\theta} = \cos \theta + j \sin \theta$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\mathbf{z} = x + jy = |\mathbf{z}|e^{j\theta}$$

$$\mathbf{z}^* = x - jy = |\mathbf{z}|e^{-j\theta}$$

$$x = \Re(\mathbf{z}) = |\mathbf{z}| \cos \theta$$

$$|\mathbf{z}| = \sqrt[+]{\mathbf{z}\mathbf{z}^*} = \sqrt{x^2 + y^2}$$

$$y = \Im(\mathbf{z}) = |\mathbf{z}| \sin \theta$$

$$\theta = \tan^{-1}(y/x)$$

$$\mathbf{z}^n = |\mathbf{z}|^n e^{jn\theta}$$

$$\mathbf{z}^{1/2} = \pm |\mathbf{z}|^{1/2} e^{j\theta/2}$$

$$\mathbf{z}_1 = x_1 + jy_1$$

$$\mathbf{z}_2 = x_2 + jy_2$$

$$\mathbf{z}_1 = \mathbf{z}_2 \text{ iff } x_1 = x_2 \text{ and } y_1 = y_2$$

$$\mathbf{z}_1 + \mathbf{z}_2 = (x_1 + x_2) + j(y_1 + y_2)$$

$$\mathbf{z}_1 \mathbf{z}_2 = |\mathbf{z}_1| |\mathbf{z}_2| e^{j(\theta_1 + \theta_2)}$$

$$\frac{\mathbf{z}_1}{\mathbf{z}_2} = \frac{|\mathbf{z}_1|}{|\mathbf{z}_2|} e^{j(\theta_1 - \theta_2)}$$

$$-1 = e^{j\pi} = e^{-j\pi} = 1 \angle \pm 180^\circ$$

$$j = e^{j\pi/2} = 1 \angle 90^\circ$$

$$-j = e^{-j\pi/2} = 1 \angle -90^\circ$$

$$\sqrt{j} = \pm e^{j\pi/4} = \pm \frac{(1 + j)}{\sqrt{2}}$$

$$\sqrt{-j} = \pm e^{-j\pi/4} = \pm \frac{(1 - j)}{\sqrt{2}}$$

**Learn how to
perform these
with your
calculator/computer**

Phasor Domain

A *domain transformation* is a mathematical process that converts a set of variables from their domain into a corresponding set of variables defined in another domain.

1. The phasor-analysis technique transforms equations from the **time domain** to the **phasor domain**.
2. **Integro-differential** equations get converted into **linear equations** with no sinusoidal functions.
3. After solving for the desired variable--such as a particular voltage or current-- in the **phasor domain**, conversion back to the **time domain** provides the same solution that would have been obtained had the original integro-differential equations been solved entirely in the time domain.

Phasor Domain

$$v(t) = V_0 \cos(\omega t + \phi) \\ = \Re[\underbrace{V_0 e^{j\phi}}_{\text{Phasor counterpart of } v(t)} e^{j\omega t}]$$

Phasor counterpart of $v(t)$

Time Domain

$$v(t) = V_0 \cos \omega t$$

$$v(t) = V_0 \cos(\omega t + \phi)$$

Phasor Domain

$$\longleftrightarrow \mathbf{V} = V_0$$

$$\longleftrightarrow \mathbf{V} = V_0 e^{j\phi}.$$

If $\phi = -\pi/2$,

$$v(t) = V_0 \cos(\omega t - \pi/2) \quad \longleftrightarrow \quad \mathbf{V} = V_0 e^{-j\pi/2}.$$

Time and Phasor Domain

$x(t)$		$\mathbf{X}$
$A \cos \omega t$	$\longleftrightarrow$	A
$A \cos(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j\phi}$
$-A \cos(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi \pm \pi)}$
$A \sin \omega t$	$\longleftrightarrow$	$Ae^{-j\pi/2} = -jA$
$A \sin(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi - \pi/2)}$
$-A \sin(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi + \pi/2)}$
$\frac{d}{dt}(x(t))$	$\longleftrightarrow$	$j\omega \mathbf{X}$
$\frac{d}{dt}[A \cos(\omega t + \phi)]$	$\longleftrightarrow$	$j\omega Ae^{j\phi}$
$\int x(t) dt$	$\longleftrightarrow$	$\frac{1}{j\omega} \mathbf{X}$
$\int A \cos(\omega t + \phi) dt$	$\longleftrightarrow$	$\frac{1}{j\omega} Ae^{j\phi}$

It is much easier to deal with exponentials in the phasor domain than sinusoidal relations in the time domain

Just need to track magnitude/phase, knowing that everything is at frequency ω

Next Lecture

- Transmission Lines Vs Free Space Propagation

